

Linux Commands — Simple Words

Quick study + practical revision guide based on the three uploaded Linux lab manuals.

1. Navigation Commands

Command	Simple meaning	Example
pwd	Shows where you are right now.	pwd
pwd -L	Shows the logical path, including symbolic-link path.	pwd -L
pwd -P	Shows the real physical path.	pwd -P
ls	Shows files and folders in the current directory.	ls
ls -l	Shows detailed information: permissions, owner, size, date, name.	ls -l
ls -a	Shows all files, including hidden files.	ls -a
ls -h	Shows file sizes in easy-to-read form such as K, M, G.	ls -lh
ls -R	Shows folders and their subfolders recursively.	ls -R
ls -t	Lists newest-modified items first.	ls -lt
cd	Goes to your home directory.	cd
cd ~	Goes to your home directory.	cd ~
cd ..	Moves one folder up.	cd ..
cd -	Goes back to the previous directory.	cd -
cd /	Goes to the root directory.	cd /

2. Directory Commands

Command	Simple meaning	Example
mkdir	Creates a new folder.	mkdir students
mkdir -p	Creates parent folders automatically when needed.	mkdir -p college/linux/week1
mkdir -m	Creates a folder with specified permissions.	mkdir -m 755 app
mkdir -v	Shows a message about each folder created.	mkdir -v first second
rmdir	Deletes an empty folder only.	rmdir students
rmdir -p	Removes an empty folder and empty parent folders.	rmdir -p parent/child
rmdir -v	Shows information while removing folders.	rmdir -v students
rmdir --ignore-fail-on-non-empty	Does not show an error when a folder is not empty.	rmdir --ignore-fail-on-non-empty folder

3. File Creation & Text Commands

Command	Simple meaning	Example
touch	Creates an empty file.	touch notes.txt
echo	Prints text on the terminal.	echo "Hello"

Command	Simple meaning	Example
echo ... >	Writes text to a file. Existing content is overwritten.	echo "Hello" > file.txt
echo ... >>	Adds text to the end of a file.	echo "World" >> file.txt

4. File Copy / Move / Delete

Command	Simple meaning	Example
cp	Copies a file.	cp a.txt b.txt
cp -r	Copies a folder and everything inside it.	cp -r dir1 dir2
cp -i	Asks before overwriting a file.	cp -i a.txt b.txt
cp -v	Shows what is being copied.	cp -v a.txt docs/
cp -u	Copies only when the source is newer.	cp -u a.txt b.txt
mv	Moves a file/folder or renames it.	mv old.txt new.txt
mv -i	Asks before overwriting.	mv -i a.txt docs/
mv -f	Forces the move/overwrite.	mv -f a.txt docs/
mv -v	Shows what is being moved.	mv -v a.txt docs/
mv -n	Does not overwrite an existing file.	mv -n a.txt docs/
rm	Deletes a file.	rm file.txt
rm -r	Deletes a folder and its contents.	rm -r folder
rm -f	Force deletes without normal prompts.	rm -f file.txt
rm -i	Asks before deleting.	rm -i file.txt
rm -d	Removes an empty directory.	rm -d folder
rm -rf	Force deletes a folder and everything inside it. USE CAREFULLY.	rm -rf folder

5. Viewing Files

Command	Simple meaning	Example
cat	Displays the whole file.	cat file.txt
cat -n	Displays the file with line numbers.	cat -n file.txt
cat -b	Numbers only non-empty lines.	cat -b file.txt
cat -E	Shows a \$ at the end of each line.	cat -E file.txt
cat -s	Reduces repeated blank lines.	cat -s file.txt
cat file1 file2	Displays two files one after another.	cat file1 file2
less	Views a file one screen at a time. Good for large files.	less file.txt
less -N	Shows line numbers while using less.	less -N file.txt
/pattern	Searches forward inside less.	/pattern
n / N	Moves to next / previous search match inside less.	n or N
q	Quits less.	q

6. Information / Help Commands

Command	Simple meaning	Example
whereis	Finds locations of a command's binary, source, and manual page.	<code>whereis gcc</code>
whereis -b	Searches only for the binary/executable.	<code>whereis -b gcc</code>
whereis -m	Searches only for the manual page.	<code>whereis -m gcc</code>
whereis -s	Searches only for source entries.	<code>whereis -s gcc</code>
whatis	Gives a short, one-line description of a command.	<code>whatis cp</code>
whatis -w	Uses wildcard matching.	<code>whatis -w 'cp*'</code>
whatis -r	Uses regular-expression matching.	<code>whatis -r 'copy.*'</code>
sudo mandb	Updates the manual-page database if <code>whatis</code> needs it.	<code>sudo mandb</code>

7. Admin / Package Commands

Command	Simple meaning	Example
sudo	Runs a command with administrator privileges.	<code>sudo apt update</code>
sudo apt update	Updates the package information list.	<code>sudo apt update</code>
sudo apt install build-essential	Installs build tools such as <code>gcc</code> , <code>g++</code> , and <code>make</code> .	<code>sudo apt install build-essential</code>

8. GCC Commands

Command	Simple meaning	Example
gcc --version	Shows the installed GCC version.	<code>gcc --version</code>
g++ --version	Shows the installed G++ version.	<code>g++ --version</code>
make --version	Shows the installed make version.	<code>make --version</code>
gcc program.c	Compiles a C program.	<code>gcc program.c</code>
gcc -E	Runs preprocessing and shows preprocessed C code.	<code>gcc -E main.c</code>
gcc -S	Produces assembly code.	<code>gcc -S main.c</code>
gcc -c	Compiles into an object file without making the final executable.	<code>gcc -c main.c</code>
gcc -o	Lets you choose the output file name.	<code>gcc -o program main.c</code>
./program	Runs a program from the current directory.	<code>./program</code>

9. Makefile Commands

Command	Simple meaning	Example
make	Reads the makefile and builds only what needs rebuilding.	<code>make</code>
rm *.o *.exe	Removes generated object files and executables.	<code>rm *.o *.exe</code>

10. GCC Compilation Flow

Remember the basic flow:

C source (.c) → Preprocessing → Compilation → Assembly → Object file (.o) → Linking → Executable

Important flags:

- **-E** = preprocessing
- **-S** = assembly output
- **-c** = create object file, do not link
- **-o** = choose output filename

11. Makefile Idea

A Makefile saves time by compiling only the files that need rebuilding.

```
main.c → main.o ■
add.c → add.o ■
sub.c → sub.o ■ → GCC → test.exe
mul.c → mul.o ■
```

Typical commands from the lab:

```
gcc -c main.c
gcc -c add.c
gcc -c sub.c
gcc -c mul.c
gcc -o test.exe main.o add.o sub.o mul.o
./test.exe
```

12. Ultra-Quick Memory Sheet

Question in your head	Command
Where am I?	pwd
What's here?	ls
Move into folder	cd
Go up one folder	cd ..
Go home	cd ~
Create folder	mkdir
Create nested folders	mkdir -p
Create empty file	touch
Write text	echo ... > file
Add text	echo ... >> file
Copy	cp
Move / rename	mv
Delete	rm
Read small file	cat
Read large file	less
Where is a command?	whereis

Question in your head	Command
What does a command mean?	whatis
Run as administrator	sudo
Compile C	gcc
Compile only to .o	gcc -c
Choose output name	gcc -o
Build using Makefile	make